



ULAB

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RESEARCH PROPOSAL

Misinformation in Science Reporting in Prominent Bangladeshi Newspapers: A Content Analysis

Submitted to	Submitted by
Nandita Tabassum Khan Senior Lecturer Department of Media Studies and Journalism University of Liberal Arts Bangladesh	Md. Miraz Hossain (233012010) Md. Mahfuzar Rahman (233012048) Md. Naif Hasan (233012007) Muntasim Hossain Adib (233012020)

1.0 Introduction

Science journalism in Bangladesh occupies an under-examined position in the country's media landscape. Newspapers remain trusted sources of information for a substantial portion of the population, and the science content they carry — on public health, environmental risk, medical developments, and technological change — directly shapes public understanding of issues that require accurate information to navigate. Whether that content is accurate, and in what ways it falls short when it is not, has received almost no systematic empirical attention in the Bangladeshi context.

This proposal addresses that gap. It outlines a content analysis of science reporting in three prominent Bangladeshi newspapers, using UNESCO's framework of seven information disorder categories as the analytical instrument. The goal is not to produce a sweeping judgement about Bangladeshi journalism but to generate a baseline empirical picture: how prevalent are different forms of misinformation in science reporting, which scientific topics are most affected, and how does the distribution of misinformation types vary across publications?

The proposal makes one conceptual argument before presenting the research design. The UNESCO framework was developed primarily with digital and social media content in mind. Applying it to print newspaper content requires some adaptation, and that adaptation is addressed explicitly in the methodology section. The seven categories — satire or parody, false connection, misleading content, false context, imposter content, manipulated content, and fabricated content — do not map onto print science journalism with equal relevance. Fabricated content and imposter content are less likely in established newspaper environments than misleading content and false connection. The coding scheme developed for this study accounts for this distribution.

1.1 Research Objectives

The study has one primary objective: to identify and categorise instances of misinformation appearing in science reporting in prominent Bangladeshi newspapers, using the UNESCO information disorder framework.

1.2 Research Questions

- What are the most prevalent categories of misinformation in science reporting across three prominent Bangladeshi newspapers?
- How does the prevalence of misinformation vary across UNESCO's seven information disorder categories?
- Are specific scientific topics or subject areas associated with higher rates of misinformation?

1.3 Scope and Limitations

The study is bounded to three newspapers, a defined three-month sample period, and English-language science content only. These constraints limit the generalisability of findings to the broader Bangladeshi newspaper landscape and to Bangla-language science reporting, which warrants separate analysis. The content analysis will identify the presence and type of misinformation but will not establish causes or trace downstream public impact — both important questions that require different methods and are beyond this study's scope.

Coding involves an inherent element of interpretive judgement, particularly at the boundary between misleading content and factually accurate but incomplete reporting. This is mitigated by a detailed coding scheme developed and piloted before main data collection, with inter-coder reliability measured on a double-coded subset. The mitigation is not elimination: the study's findings should be read as an evidence-based estimate, not a definitive audit.

2.0 Literature Review

The literature on misinformation in science reporting is concentrated in three areas: definitional and measurement debates, empirical studies of misinformation prevalence in Western media contexts, and a smaller body of work on the specific conditions that make science journalism vulnerable to inaccuracy. This review moves through those three areas before establishing the specific gap this study addresses.

2.1 Defining and Measuring Scientific Misinformation

Southwell et al. (2022) provide the most operationally useful definition for this study: scientific misinformation is publicly available information that is misleading or deceptive relative to the best available scientific evidence, contrary to the statements of

actors or institutions adhering to scientific principles. Crucially, they argue that scientific misinformation is observable and measurable — a claim that is sometimes contested — but that research in this area frequently misses opportunities to define clear units of analysis and to engage practising scientists in the classification process. Their methodological observations directly inform the coding scheme this study uses.

Osman et al. (2022), working from a large-scale survey, find that there is substantial public disagreement about what misinformation means, particularly regarding intentionality. Most survey respondents defined misinformation as information “intentionally designed to mislead,” which would exclude honest errors and well-intentioned oversimplifications. The UNESCO framework takes a different position, treating the information disorder categories as descriptive rather than intentional — a piece of content can be misleading without the journalist having intended to mislead. This study follows the UNESCO approach: the analysis codes what the content does, not what the journalist intended.

2.2 Misinformation in Science Journalism: Empirical Evidence

The empirical record on misinformation in science journalism is uneven but consistent in one direction. Jaafar and Giam (2012) found that only 46.6% of online news reports on a scientific discovery were factually accurate in a sample of conservation biology coverage — a finding that, while specific to one domain, established that inaccuracy rates in science journalism can be substantially higher than practitioners tend to assume. Dempster, Sutherland, and Keogh (2022), examining news media coverage of scientific research more broadly, identified three recurring patterns: misrepresentation of statistical findings, sensationalist framing that overstated the significance of preliminary results, and recommendations that the underlying research did not support. These patterns map onto the UNESCO categories of misleading content and false connection more than onto fabrication or manipulation.

Chan and Albarraciún (2023), in a meta-analysis of correction effects in science misinformation, find that corrections reduce belief in misinformation but do not eliminate it, and that the residual belief effect varies significantly by topic and by the prior beliefs of the audience. This has direct relevance for why measuring the type and

prevalence of misinformation in science reporting matters: different types of misinformation are likely to require different corrective approaches, and the distribution of types in a given media environment shapes what correction strategies would be most effective.

2.3 The Bangladeshi Context

Peer-reviewed empirical research on science journalism specifically in Bangladesh is essentially absent from the literature. The closest available evidence comes from health journalism. Moshir Rahman Khasru et al. (2021), in a pilot study of newspaper health reporting, found that only 44% of health articles had a positive orientation toward health services, with a significant proportion related to malpractice or treatment errors. More relevant for present purposes: the study documented a shortage of medical and scientific knowledge among health journalists, which the authors attributed to the absence of specialist training in science communication and the general reliance on press releases and official statements as primary sources. These structural conditions — limited specialist knowledge, press-release dependence, absence of formal training in evaluating scientific evidence — create the conditions under which the forms of misinformation Dempster et al. identify are most likely to appear.

Pseudoscience as a distinct category of science misinformation is also relevant in the Bangladeshi context. Pigliucci and Boudry (2013) draw the necessary distinction between legitimate scientific uncertainty, which is a feature of science rather than a flaw, and pseudoscientific claims that reject the standards of empirical evidence altogether. Bangladeshi newspapers cover topics — alternative medicine, environmental health, climate-adjacent claims — where this distinction is practically important and where the UNESCO framework's misleading content and false context categories would be most commonly invoked.

2.4 Addressing Misinformation: Implications for Intervention

Stoltman et al. (2023) propose, in the context of addiction journalism, a collaborative approach involving academics, journalists, and community organisations in developing shared standards for evidence-based reporting. The model is adaptable: what it requires is subject-matter expertise feeding into reporting practice, rather than

expertise sitting entirely outside the newsroom. Kazemi and Mihalcea (2023) frame misinformation as “information pollution” and argue that civil society actors, journalists, and fact-checkers have distinct and complementary roles in reducing its prevalence. Both frameworks suggest that the most productive response to science misinformation in journalism is structural rather than individual — training, editorial standards, and institutional access to scientific expertise rather than correction of individual articles after publication.

2.5 Gap This Study Addresses

The literature establishes three things relevant to this proposal. First, misinformation in science journalism is a documented and measurable phenomenon with specific recurring forms. Second, the structural conditions in Bangladeshi newsrooms — limited scientific expertise, press-release dependence, absence of specialist training — are the conditions associated with the highest misinformation risk in the literature. Third, no published study has examined science misinformation specifically in Bangladeshi print media using a systematic content analysis methodology.

The present study is a direct attempt to address the third of these gaps, with awareness of the first two shaping the coding scheme and the interpretation of findings.

3.0 Research Design

3.1 Research Methodology

The study uses quantitative content analysis. This choice is driven by the research questions: they ask about prevalence and distribution across categories, which requires systematic coding of a defined corpus. Content analysis allows for objective coding using a specified scheme, quantification of misinformation type frequencies, and comparison across publications and scientific topic areas. Its limitation in this context is that it does not establish why misinformation appears where it does — that would require interviews with journalists and editors, which are beyond this study’s scope but represent the natural next step from its findings.

The analytical framework is UNESCO's seven-category information disorder typology: satire or parody, false connection, misleading content, false context, imposter content, manipulated content, and fabricated content. The framework is applied with two adaptations for print science journalism. First, satire or parody is retained as a category but is expected to appear rarely in the target publications. Second, a category distinction is maintained between misleading content that results from active framing choices (sensationalism, selective evidence) and misleading content that results from omission (leaving out relevant caveats or context). This sub-distinction does not add a new category to the UNESCO scheme but adds a coding variable within the misleading content category that the original framework does not specify.

3.2 Sampling

Three prominent Bangladeshi English-language newspapers are selected: The Daily Star, The Business Standard, and Prothom Alo's English digital edition. Selection criteria are circulation, reputation for institutional credibility, and the presence of identifiable science or health sections. English-language selection is a scope constraint, not a claim that English-language newspapers are more important than Bangla-language ones; it reflects the practical constraint of coding scheme validity across languages.

The sampling period covers three months selected to capture different news cycles: March 2020 (COVID-19 initial reporting period, high-volume science content), January 2021 (vaccine rollout reporting), and January 2022 (Omicron wave coverage). These periods are selected deliberately rather than randomly because they represent moments of high public engagement with science content and therefore higher stakes for accuracy. The limitation is that findings from these periods may not be representative of routine science reporting in lower-stakes periods.

Within each month, all articles tagged under science, health, environment, or technology sections are included in the corpus. Articles that reference scientific claims incidentally (a political story that mentions a health statistic) are excluded. The boundary is defined as articles where a scientific claim or finding is a primary subject of the reporting.

3.3 Coding and Inter-Coder Reliability

The coding scheme is developed through three steps: initial scheme construction based on the UNESCO categories and the literature review, pilot coding of a 10% random sample of the corpus by two coders independently, and revision of the scheme based on disagreements identified in the pilot. Main coding proceeds after the revised scheme achieves Cohen's $\kappa \geq .70$ on the pilot sample. All disagreements in main coding are resolved through discussion and documented. The final reliability figure is reported in the study.

3.4 Timeframe

Phase	Activity	Duration
1	Instrument development and ethics approval	15 days
2	Literature review (finalisation)	20 days
3	Data collection and corpus construction	30 days
4	Pilot coding and scheme revision	15 days
5	Main coding	30 days
6	Analysis and write-up	30 days
	Total	140 days (approx. 4.7 months)

3.5 Budget

Category	Amount (BDT)
Resources (journals, database access, publications)	40,000
Stationery and printing	1,000
Equipment and internet	1,50,000
Transport	20,000
Research assistant remuneration	50,000
Miscellaneous	10,000
Total	2,71,000

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